Building a Machine Learning Classifier Using Fundamental Financial Ratios for Long-Term Equity Selection

1. Introduction and Problem Statement

Equity markets are influenced by a wide range of parameters-long term growth trends, sentiments and random noise. As written in “The Intelligent Investor” what prevails above this all is long term investment- not trading on random noise. Investing in companies with truly good fundamental value will help us ensure growth in our portfolio without needing the high frequency trading that comes with other strageies. Investing in true value- at a reasonable price can give us a return far greater than keeping our money idle in any bank without having to slave at it every day. ML seems like the perfect solution in all of this- it will help us determine which rations and other balance sheet and PnL items actually are linked with price growth- assuming that any company with long term price growth is growing due to its strength in fundamentals.

The objective of this project is to explore whether machine learning models trained on fundamental financial ratios can assist in making long-term investment decisions. Unlike algorithmic trading, which focuses on predicting short-term price fluctuations using high-frequency data, this project restricts itself to long-term investment outcomes based on company fundamentals. The problem is framed as a classification task, where equities are classified as either successful or unsuccessful investments over a long-term horizon.

Success is defined as a price appreciation greater than 8%, a threshold chosen to represent meaningful long-term value creation beyond inflation and low-risk benchmark returns. By framing the problem this way, the project aims to align machine learning predictions with realistic investment objectives rather than speculative short-term gains.

The project investigates three core questions:

1. Can fundamental financial ratios be used effectively as features for machine learning-based equity classification?
2. How do different machine learning models perform on this task, and what are the trade-offs between interpretability and predictive power?

3. How do machine learning-based predictions compare with conventional methods of equity selection, and what are the implications for real-world investing?

2. Financial Ratios and Conventional Equity Selection

Financial ratios are quantitative measures derived from a company's financial statements and are central to conventional equity analysis. These ratios provide insights into profitability, valuation, risk, efficiency, and financial stability. Investors and analysts commonly rely on these ratios to compare companies within and across industries.

Some of the most widely used categories of financial ratios include:

- **Profitability Ratios** such as Return on Equity (ROE) and Return on Assets (ROA), which measure how efficiently a company generates profits from its capital and assets.
- **Valuation Ratios** such as Price-to-Earnings (PE) and Price-to-Book (PB), which indicate how the market values a company relative to its earnings or book value.
- **Risk and Volatility Measures** such as Beta, which capture a stock's sensitivity to overall market movements.
- **Growth and Leverage Ratios**, which provide insight into a company's expansion potential and financial risk.

We worked on this in our first week- we were tasked to go through the zerodha modules, especially the fundamental analysis ones. These really helped me revise my pre-existing knowledge in this domain while also teaching me new things. I was used to just seeing various ratios as just numbers with arbitrary figures in my brain as to what makes a ration good or bad. These modules walked us through how each ratio was calculated and how the key of fundamental analysis is industry wise comparison as these ratios greatly differ from sector to sector. It also taught me a key concept- valuation, especially the DCF analysis.

Conventional equity selection often involves analyzing these ratios individually or in small groups, sometimes supplemented by valuation models such as Discounted Cash Flow (DCF) analysis. DCF attempts to estimate a company's intrinsic value by forecasting future cash flows and discounting them back to the present. While conceptually sound, DCF models are highly sensitive to assumptions about growth rates, discount rates, and terminal values, making them subjective and prone to bias.

Although traditional fundamental analysis is interpretable and grounded in economic intuition, it has several limitations. Human analysts may struggle to process large numbers of interacting variables simultaneously, often relying on linear reasoning or heuristics. This limitation motivates the use of machine learning, which can model complex, non-linear relationships across many financial ratios simultaneously.

3. Conceptualising ML

During the initial phase of the project, emphasis was placed on understanding core machine learning concepts such as bias-variance tradeoff, overfitting, and generalization. These concepts are particularly relevant in financial datasets, which are often noisy, limited in size, and characterized by correlated features.

High-bias models may oversimplify the relationship between financial ratios and long-term returns, failing to capture meaningful patterns. Conversely, high-variance models may fit historical data extremely well but perform poorly on unseen data, a common risk when modeling financial markets.

This learning phase highlighted the importance of using multiple models with varying complexity. Simpler models such as Logistic Regression provide interpretability and serve as strong baselines, while more complex models such as Random Forests can capture non-linear interactions at the cost of reduced transparency. Understanding these trade-offs informed later modeling decisions in the project.

For someone who has never dealt with ML before, going through the chapters provided(excerpts from “Introduction to Statistical Learning” by James Witten) helped me understand what ML was beyond buzzwords and jargon. We’re essentially figuring out the mathematical relation between our features and modelling so we can predict outputs as required.

Overfitting which is very easy especially in financial models- how it should be avoided. The types of bias and how they should be avoided. The trade off between variance, bias etc. and the different types of models and how they worked helped me cement the idea of ML into my head from a far away idea into an actual mathematical model.

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4. Manual Ratio Computation and Financial Statement Analysis

To develop a deeper understanding of the data used in the machine learning pipeline, financial ratios were manually computed for a selected company using its annual report. This exercise required analyzing balance sheets, income statements, and cash flow statements in detail.

Manually computing ratios served us two key purposes. First, it reinforced the accounting logic behind commonly used financial metrics, highlighting how managerial decisions and accounting practices influence reported numbers. It helped give us a feel on how various metrics are related to each other. Second, it emphasized the importance of understanding the data generation process, particularly when using scraped or aggregated financial datasets.

This exercise also improved the ability to interpret model outputs later in the project. By understanding how ratios are constructed and what they represent economically, model predictions could be evaluated for financial plausibility rather than treated as abstract numerical outputs.

I chose the stock Cello world. I extracted the PnL, balance sheet and cash flows from the annual report and calculated all the rations necessary manually while also performing dcf to figure out the integral value of the company. This gave me insight into how fundamental analysis really works and what mindframe real analysts are in when investigating stocks.

The excel sheet: wids_assignment1-1

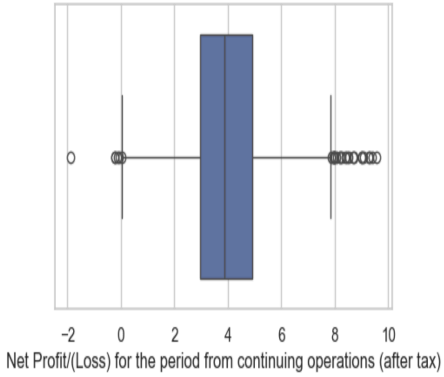
5. Exploratory Data Analysis

Exploratory Data Analysis (EDA) was conducted to understand the distribution, scale, and quality of the dataset. Financial ratios were observed to exhibit significant skewness and the presence of extreme values, which is common in financial data due to differences in firm size, profitability cycles, and industry structure.

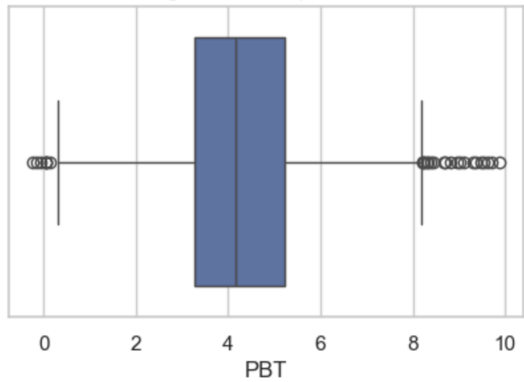
Exploratory data analysis helps us understand what our dataset is and how we can modify it to serve our needs. It gives us a general idea of the values of our data set, how they're distributed and what our missing values are.

Instead of normal box plots- log scaled box plots help us understand the trends more. Normal box plots are extremely skewed, not giving us any feel of how the data actually is. Following are some interesting log scaled box plots:

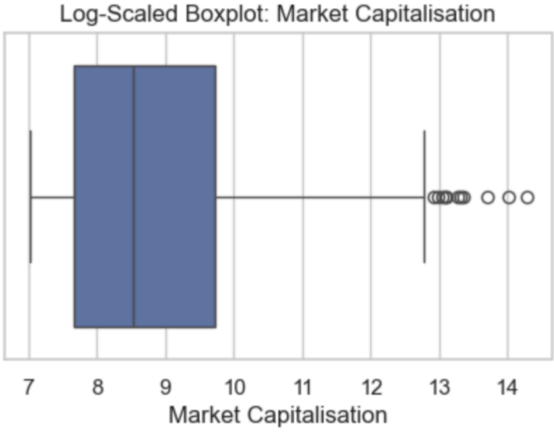
Log-Scaled Boxplot: Net Profit/(Loss) for the period from continuing operations (after tax)



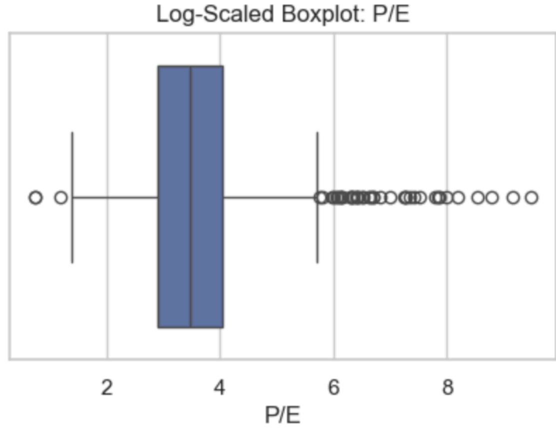
Log-Scaled Boxplot: PBT



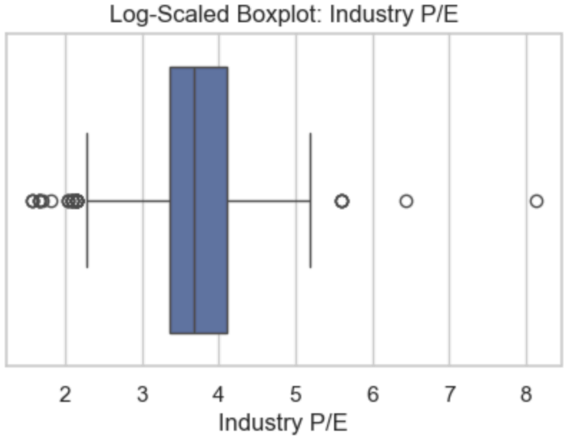
Log-Scaled Boxplot: Market Capitalisation



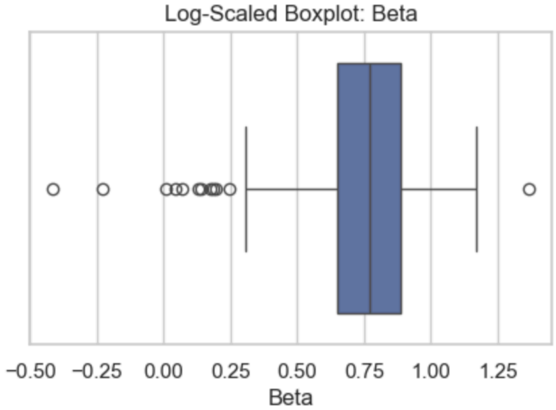
Log-Scaled Boxplot: P/E



Log-Scaled Boxplot: Industry P/E



Log-Scaled Boxplot: Beta



Visualizations such as histograms and box plots revealed that several valuation ratios contained outliers, while others showed long-tailed distributions. These observations influenced later preprocessing decisions, particularly the choice of median-based imputation for skewed variables.

EDA also helped identify missing values across features. Rather than applying a uniform strategy, the proportion and nature of missingness were analyzed for each variable, leading to differentiated imputation strategies tailored to financial context.

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6. Data Preprocessing and Feature Engineering

Data preprocessing was one of the most critical stages of the project.

Feature	Missing %
Beta	11.72
Change in stock	30.66
P/E	10.42
Total tax provision	7.11
Depreciation	6.01
Interest expenses	4.61
NSE symbol	1.2

Feature	Missing %
PBT	0.4
Raw materials, stocks, spares, purchase of fini...	30.06
Reported Profit after tax	0.4
Total expenses	0.4
Total income from continuing operations	0.4
Date	0.4
Industry group	0

Feature	Missing %
Net Profit/(Loss) from continuing operations	1.2
Earnings per share before extraordinary item	0.9
Enterprise value	0.8
Net sales	0.7
PBDT	0.4
PBIT net of P&E	0.4
PBDIT	0.4
PBT	0.4
Reported Profit after tax	0.4
Total expenses	0.4

Feature	Missing %
Closing Price	0
Date (duplicate)	0
Closing Price (duplicate)	0
Shares Outstanding	0
Market Capitalisation	0
EPS	0
Number of Transactions	0
ISIN code	0
Industry P/E	0
Company Name	0

Features with greater than 30% missing values were dropped. Imputing them to any amount would involve a great amount of assumption about the distribution of data and could cause our ML model to pick up on variances that do not exist. Hence rather than using these features to train our model they were dropped all together.

For features such as PE ratio and Beta, which had approximately 10% missing data, imputation methods were chosen to preserve the underlying distribution. This approach ensured that key valuation and risk characteristics were retained without artificially compressing variability.

Other numerical features with relatively few missing values were imputed using median values. Median imputation was preferred due to the skewed nature of many financial ratios and its robustness to outliers.

An industry-level feature was also engineered to capture sectoral effects. This feature combined industry PE and market capitalization information and was normalized to produce a composite industry score. This approach allowed sector characteristics to be incorporated into the model without relying on high-dimensional one-hot encoding, which can be inefficient and noisy in financial datasets.

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7. Model Building and Threshold Definition

Multiple machine learning models were trained and evaluated, including Logistic Regression and Random Forest classifiers.Classifiers were preferred over models predicting f(x) which could also be implemented. But our main purpose of ML is to simply define if our model is “investable” or not. Hence Logistic Regression served as an interpretable baseline, allowing examination of how individual ratios influenced investment outcomes.

Given the long-term investment focus, a classification threshold of price gain greater than 8% was chosen to define successful investments. This threshold reflects a meaningful return level that exceeds low-risk alternatives and accounts for long-term investment horizons. This 8% is a low enough baseline that accounts for cost of fund and some depreciation. Ideally I would like it to be higher-but given the dataset and the years I’m using I went with a lower number.

Model evaluation emphasized not only predictive performance but also stability and resistance to overfitting. Train-test splits were used to assess generalization, and model behavior was analyzed for financial consistency.

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8. Hybrid Modeling Approach

While Logistic Regression provided interpretability, it was limited in capturing complex non-linear interactions between financial ratios. Random Forest models addressed this limitation by modeling higher-order interactions but introduced risks of overfitting and reduced transparency.

To balance these trade-offs, outputs from Logistic Regression and Random Forest models were combined to form the final machine learning framework. This hybrid approach leveraged the stability and interpretability of linear models while incorporating the expressive power of tree-based methods.

The final model selection prioritized robustness and financial plausibility over marginal gains in accuracy.

Below is the report for a random forest classifier.

Class	Precision	Recall	F1-score	Support
Not Investable	0.77	0.95	0.85	144
Investable	0.67	0.25	0.36	56
Accuracy	—	—	0.76	200
Macro Avg	0.72	0.6	0.61	200
Weighted Avg	0.74	0.76	0.71	200

In the above random forest when I treated both investable and non investable as the same it gave really poor recall as seen above: i.e. missed many investment opportunities. In my strategy missing investment opportunities should be more costly than missing losses- as losses can be earnt back but the true way to play the market is investing at times when others don't see.

Logistic regression gave better results- even better than random forest and logistic combined.Hence in the end I went with that.

In the logistic regression I tried optimising c but that again led to poor accuracy which might have been due to overfitting. My final model was just a logistic regression which performed best for this data.

Results of that are given below:

Class	Precision	Recall	F1-Score	Support
0	0.85	0.88	0.86	100
1	0.88	0.84	0.86	100
Accuracy	—	—	0.86	200
Macro Avg	0.86	0.86	0.86	200
Weighted Avg	0.86	0.86	0.86	200

The above classification report reveals that the model is reliable with equal balances of false positives and false negatives(accuracy was deteiorating too much if I favoured one).

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9. Comparison with Conventional Equity Selection

Compared to traditional equity selection methods, the machine learning approach demonstrated strengths in scalability and systematic pattern recognition. While conventional analysis excels in interpretability and contextual judgment, it is limited in handling large numbers of interacting variables simultaneously.

Machine learning models, when properly constrained and interpreted, can act as decision-support tools that complement human judgment rather than replace it. This project demonstrates that ML-based classification can augment traditional analysis by highlighting patterns that may not be immediately apparent through manual ratio inspection.

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10. Conclusion, Limitations, and Future Work

This project explored the application of machine learning to long-term equity selection using fundamental financial ratios. The results suggest that ML models can provide useful insights when grounded in financial understanding and supported by robust preprocessing.

However, the study has limitations, including limited historical scope, sensitivity to label definition, and the exclusion of macroeconomic variables. Future work could incorporate time-series modeling, sector-specific models, and macro-fundamental integration.

Overall, the project reinforces the idea that machine learning is most effective in finance when used as a complement to, rather than a substitute for, conventional financial analysis.